

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Lakshana K [192524433] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. A media company wants to migrate its video streaming platform to the cloud.

Answer

A media company requires a cloud architecture that can support millions of users simultaneously while providing uninterrupted video streaming. A scalable cloud architecture should include multiple application servers, load balancers, content delivery networks (CDNs), caching services, and cloud storage.

The video files are stored in cloud object storage, while frequently accessed videos are cached using a CDN. The CDN stores copies of videos at locations closer to users, reducing buffering and improving streaming speed. Application servers process user requests, while databases store user accounts and viewing history.

Load balancing distributes incoming traffic evenly across multiple servers. If one server becomes overloaded, requests are automatically redirected to other available servers. This improves performance and prevents server failures.

Auto-scaling automatically increases the number of servers during peak hours and decreases them during low traffic periods. This ensures consistent performance while reducing unnecessary resource usage.

Cost Optimization Strategies

- Use Auto Scaling to avoid paying for unused resources.
- Store old videos in low-cost cloud storage.
- Use CDN caching to reduce bandwidth costs.
- Monitor resource usage and remove idle instances.
- Use Reserved or Spot Instances where suitable.

Conclusion

A combination of CDN, load balancing, auto-scaling, and cloud storage ensures high performance, minimal buffering, and cost-effective video streaming.

2. A fintech startup is deploying applications using containers across multiple cloud providers.

Answer

A fintech startup can use **Docker containers** to package applications and **Kubernetes** for container orchestration across multiple cloud providers. Kubernetes ensures that applications run consistently regardless of the cloud platform.

Container orchestration automates deployment, scaling, monitoring, and recovery of containers. If one container fails, Kubernetes automatically replaces it, ensuring continuous service.

Service discovery allows containers to locate each other automatically using Kubernetes Services and DNS. This eliminates the need for manual configuration.

Scaling is handled through Kubernetes Horizontal Pod Autoscaler, which increases or decreases container instances depending on CPU or memory usage.

Benefits of Multi-Cloud

- Higher availability.
- Avoids vendor lock-in.
- Better disaster recovery.
- Improved performance by choosing the best cloud provider.

Risks

- Increased management complexity.
- Higher networking costs.
- Security policy differences.
- More complex monitoring.

Conclusion

Using Kubernetes with a multi-cloud strategy provides flexibility, reliability, and scalability while ensuring consistent deployments.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.

Answer

The e-learning platform should implement a cloud-native backup and disaster recovery solution to protect important student records, course materials, and user data.

Automatic backups should be scheduled daily, while critical databases should use continuous replication to another cloud region. Object storage versioning should be enabled so that deleted or modified files can be restored easily.

Cloud snapshots should be created regularly for virtual machines and databases. Backup copies should be stored in different geographic regions to protect against regional failures.

Recovery Time Objective (RTO) is achieved by keeping standby systems ready so applications can restart quickly.

Recovery Point Objective (RPO) is achieved through frequent backups and real-time replication, minimizing data loss.

Business Continuity Steps

- Enable automatic backups.
- Store backups in multiple regions.
- Test disaster recovery regularly.
- Monitor backup status continuously.
- Prepare a disaster recovery plan.

Conclusion

Backup, replication, and versioning provide reliable protection against accidental deletion while ensuring quick recovery and continuous learning services.

4. A government agency is adopting a hybrid cloud model.

Answer

A hybrid cloud combines on-premises infrastructure with public cloud services. Sensitive government applications remain on-premises to satisfy compliance requirements, while non-sensitive applications such as public websites and document portals run in the cloud.

Secure communication between environments is established using VPN or dedicated private connections. Firewalls and network security groups protect data exchanged between cloud and on-premises systems.

Identity Management

A centralized Identity and Access Management (IAM) system provides Single Sign-On (SSO), Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC) to ensure only authorized users can access resources.

Encryption

Sensitive data should be encrypted during transmission using TLS and at rest using AES-256 encryption. Encryption keys should be managed securely through cloud key management services.

Challenges

- Managing multiple environments.
- Maintaining consistent security policies.

- Compliance and regulatory requirements.
- Higher operational complexity.
- Increased monitoring requirements.

Conclusion

A hybrid cloud offers flexibility while maintaining security and compliance, making it suitable for government organizations.

5. A SaaS provider needs high availability and fault tolerance.

Answer

A SaaS provider should deploy its application across multiple cloud regions. Each region contains application servers, databases, load balancers, and storage services.

Traffic is distributed using Global Load Balancers. If one region becomes unavailable, traffic is automatically redirected to another healthy region without affecting users.

Database replication is configured between regions using synchronous or asynchronous replication. Regular backups ensure data protection even during failures.

Failover Mechanism

Health checks continuously monitor application availability. If a region fails, automatic failover redirects users to another active region with minimal downtime.

Monitoring and Alerting

- Monitor CPU, memory, and network usage.
- Track application response time.
- Configure alerts for failures.
- Monitor database replication.
- Use centralized logging and dashboards.

Conclusion

A multi-region deployment combined with replication, failover, monitoring, and automated alerts provides high availability, fault tolerance, and uninterrupted service for SaaS applications.

